

DNA and RNA

Become familiar with the basic structures and properties of DNA, RNA. We will come back to these many times during the course. All the machineries of Molecular Biology operate on these molecules. They are the foundation.

Reading:

MBoC(6th) Ch2: *A Survey of the Nucleic Acids*, Pg. 100-101. **Ch4:** *DNA, Chromosomes and Genes*. Pg. 173-181, 187-193, 214-215. **Ch5:** 251-253.

Need to know and understand

Monomers

Bases - Purines (Adenine and Guanine) and Pyrimidines (Cytosine, Thymine and Uracil).

Know that there are many more.

nucleoside = base + sugar (ribose or deoxyribose)

nucleotide = nucleoside + (1 or 2 or 3) phosphate(s)

In DNA bases are A, C, G and T and sugar is deoxyribose.

In RNA bases are A, C, G and U and sugar is ribose

learn naming conventions. NTP, dNTP, NMP, dNMP etc.

Polymers

Unit is nucleoside monophosphate.

Phosphodiester bonds (5' of sugar:phosphate:3' of next sugar)

Polymer is directional. It has a 5' end and a 3' end. There is no limit to number of monomers.

Genetic information is stored in the sequence of bases along the length of the polymer.

Double stranded

ssDNA = single-stranded DNA

dsDNA = double-stranded DNA

each strand may be either DNA or RNA

hydrogen bonding - bases pair (bp): A with T, A with U, C with G. Notice purine with pyrimidine

base stacking

base pairing and stacking constrain bond rotations and makes the double-stranded molecule rigid.

Forms a right-handed double helix with a diameter of 2 nm. In the relaxed state there are approx. 10 bases/turn. The two sugar phosphate backbones are not directly opposite each other, so there is a major groove and a minor groove.

DNA denaturation/reassociation. Melting temperature - dependence on DNA length, CG content, salt concentration.

Supercoiling. >10 bp/turn = negative supercoil (underwound), <10bp/turn = positive supercoil (overwound).

Topoisomerases can change supercoiling by cutting one strand of DNA and letting it rotate around the other strand (Topoisomerase type 1) or, where one piece of dsDNA crosses another, by cutting both strands of one piece and passing the other piece from one side to the other (Topoisomerase type 2).

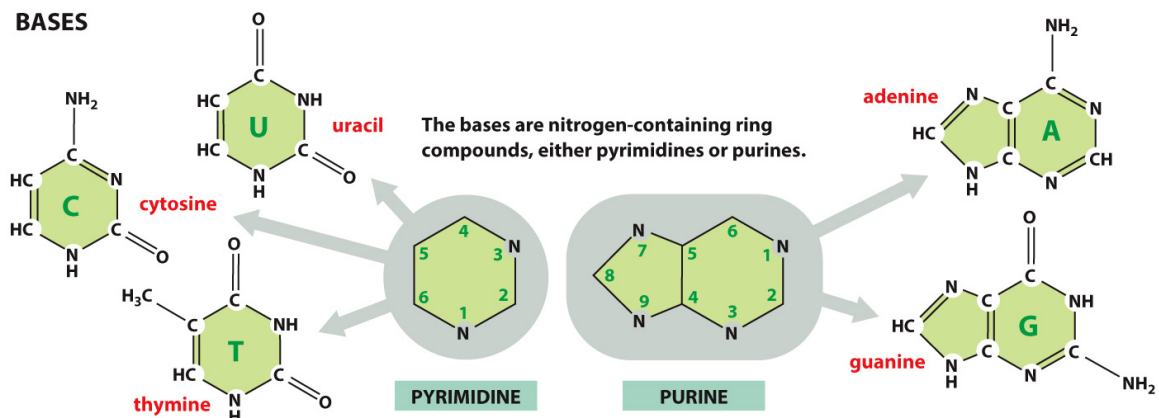
Genome

All the DNA of the organism. It is divided into chromosomes. Bacteria have one, circular chromosome. Eukaryotes have multiple linear chromosomes that are present in pairs in the diploid phase of the lifecycle and in one copy in the haploid phase.

Chromatin

Eukaryotes only! A nucleosome: DNA is wound (left-handed!) 1.7x (150 bp) around an octamer core of 8 histones 2x(H2A, H2B, H3, H4). The complex of DNA and protein is chromatin.

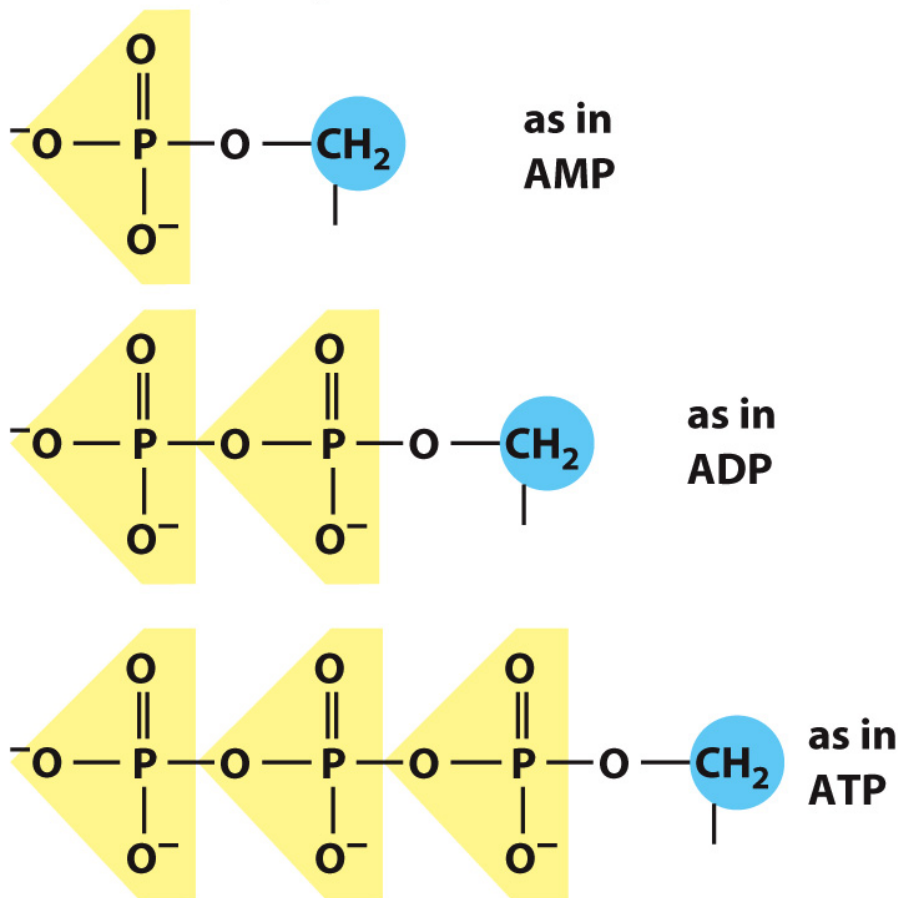
DNA can be variably packed up. The loosest is called euchromatin, which is DNA wound around nucleosomes unattached to each other. When it is packaged more tightly it is called heterochromatin, H1 is involved in this and binds adjacent nucleosomes together. At its most compact, it forms visible bodies during cell division, the metaphase chromosomes.



Panel 2-6 (Part 1) Molecular Biology of the Cell 6e (© Garland Science 2015)

PHOSPHATES

The phosphates are normally joined to the C5 hydroxyl of the ribose or deoxyribose sugar (designated 5'). Mono-, di-, and triphosphates are common.

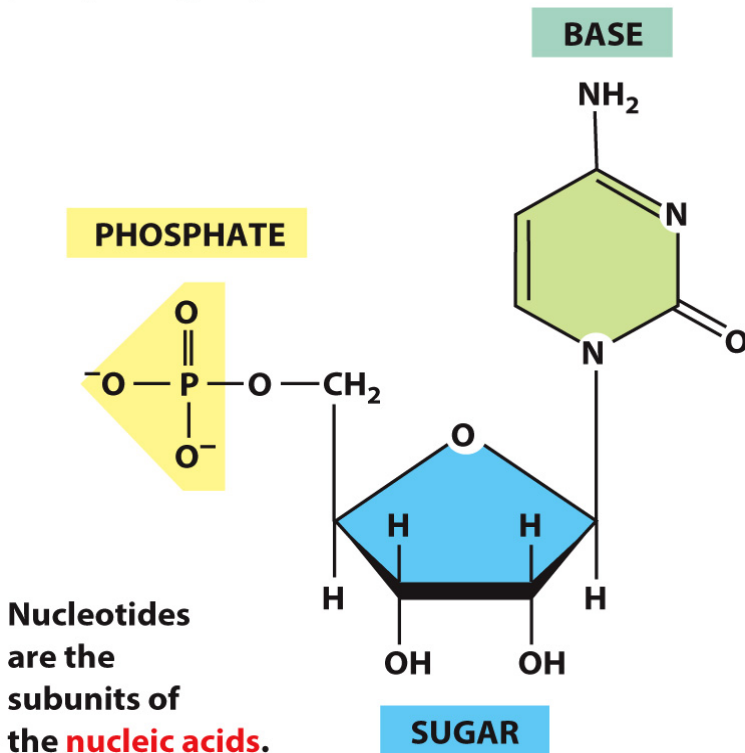


The phosphate makes a nucleotide negatively charged.

Panel 2-6 (Part 2) Molecular Biology of the Cell 6e (© Garland Science 2015)

NUCLEOTIDES

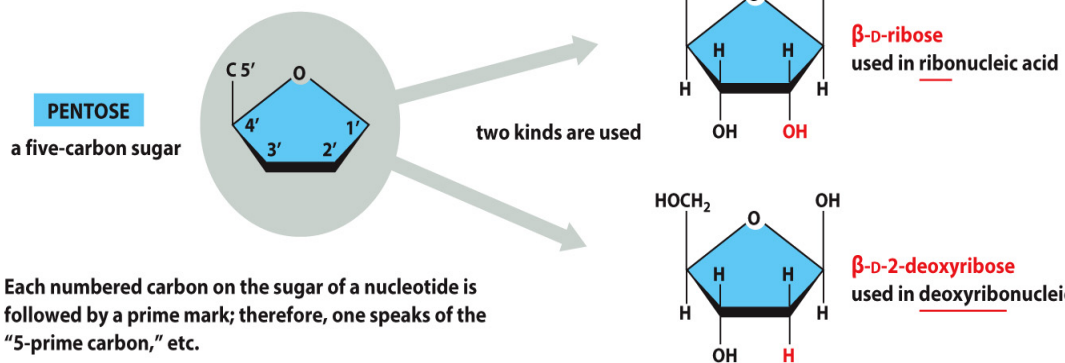
A nucleotide consists of a nitrogen-containing base, a five-carbon sugar, and one or more phosphate groups.



Nucleotides are the subunits of the **nucleic acids**.

Panel 2-6 (Part 3) Molecular Biology of the Cell 6e (© Garland Science 2015)

SUGARS



Each numbered carbon on the sugar of a nucleotide is followed by a prime mark; therefore, one speaks of the "5-prime carbon," etc.

Panel 2-6 (Part 5) Molecular Biology of the Cell 6e (© Garland Science 2015)

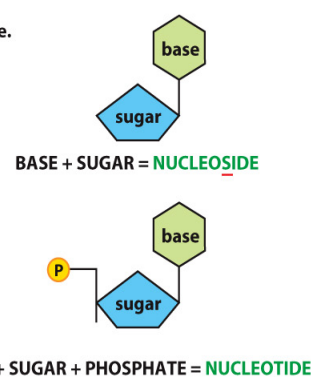
NOMENCLATURE

A nucleoside or nucleotide is named according to its nitrogenous base.

BASE	NUCLEOSIDE	ABBR.
adenine	adenosine	A
guanine	guanosine	G
cytosine	cytidine	C
uracil	uridine	U
thymine	thymidine	T

Single-letter abbreviations are used variously as shorthand for (1) the base alone, (2) the nucleoside, or (3) the whole nucleotide—the context will usually make clear which of the three entities is meant. When the context is not sufficient, we will add the terms "base", "nucleoside", "nucleotide", or—as in the examples below—use the full 3-letter nucleotide code.

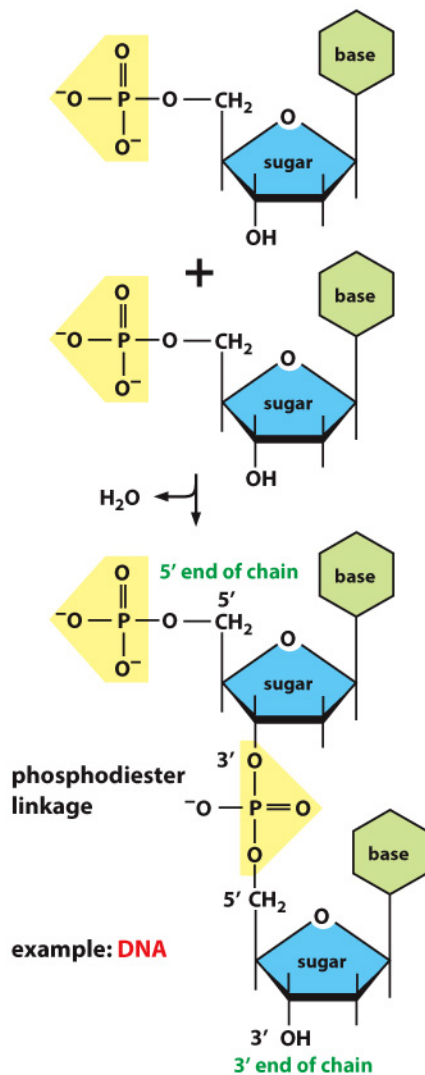
AMP = adenosine monophosphate
dAMP = deoxyadenosine monophosphate
UDP = uridine diphosphate
ATP = adenosine triphosphate



Panel 2-6 (Part 6) Molecular Biology of the Cell 6e (© Garland Science 2015)

NUCLEIC ACIDS

Nucleotides are joined together by a **phosphodiester linkage** between 5' and 3' carbon atoms to form nucleic acids. The linear sequence of nucleotides in a nucleic acid chain is commonly abbreviated by a one-letter code, such as A—G—C—T—T—A—C—A, with the 5' end of the chain at the left.



Panel 2-6 (Part 7) Molecular Biology of the Cell 6e (© Garland Science 2015)

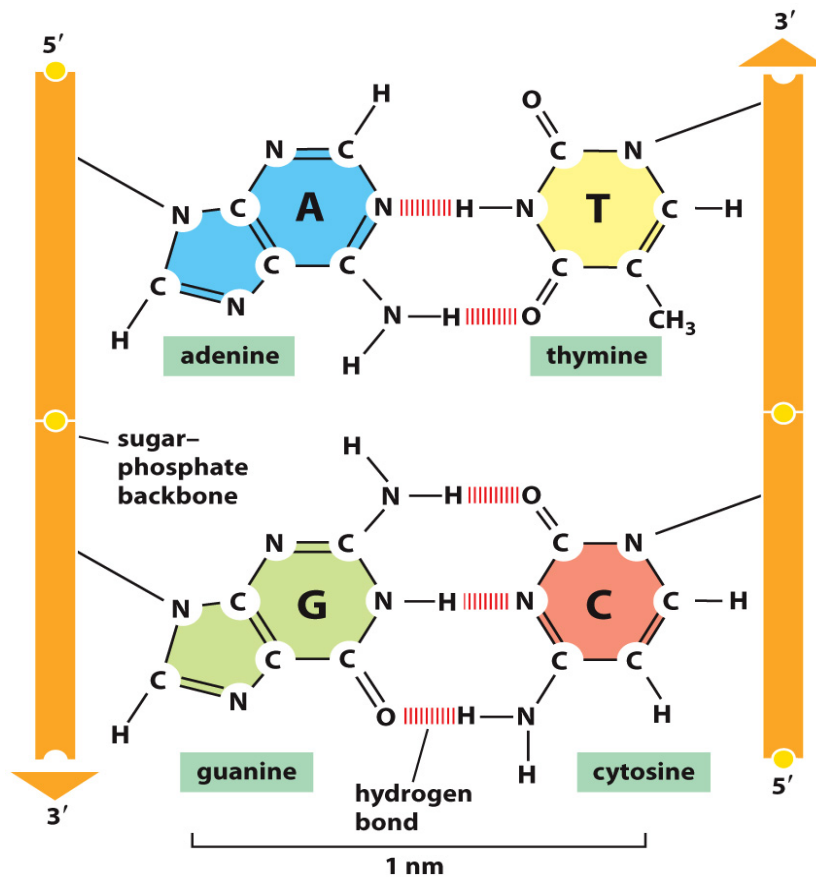


Figure 4-4 Molecular Biology of the Cell 6e (© Garland Science 2015)

Figure 4-4 Complementary base pairs in the DNA double helix. The shapes and chemical structures of the bases allow hydrogen bonds to form efficiently only between A and T and between G and C, because atoms that are able to form hydrogen bonds (see Panel 2-3, pp. 94-95) can then be brought close together without distorting the double helix. As indicated, two hydrogen bonds form between A and T, while three form between G and C. The bases can pair in this way only if the two polynucleotide chains that contain them are antiparallel to each other.

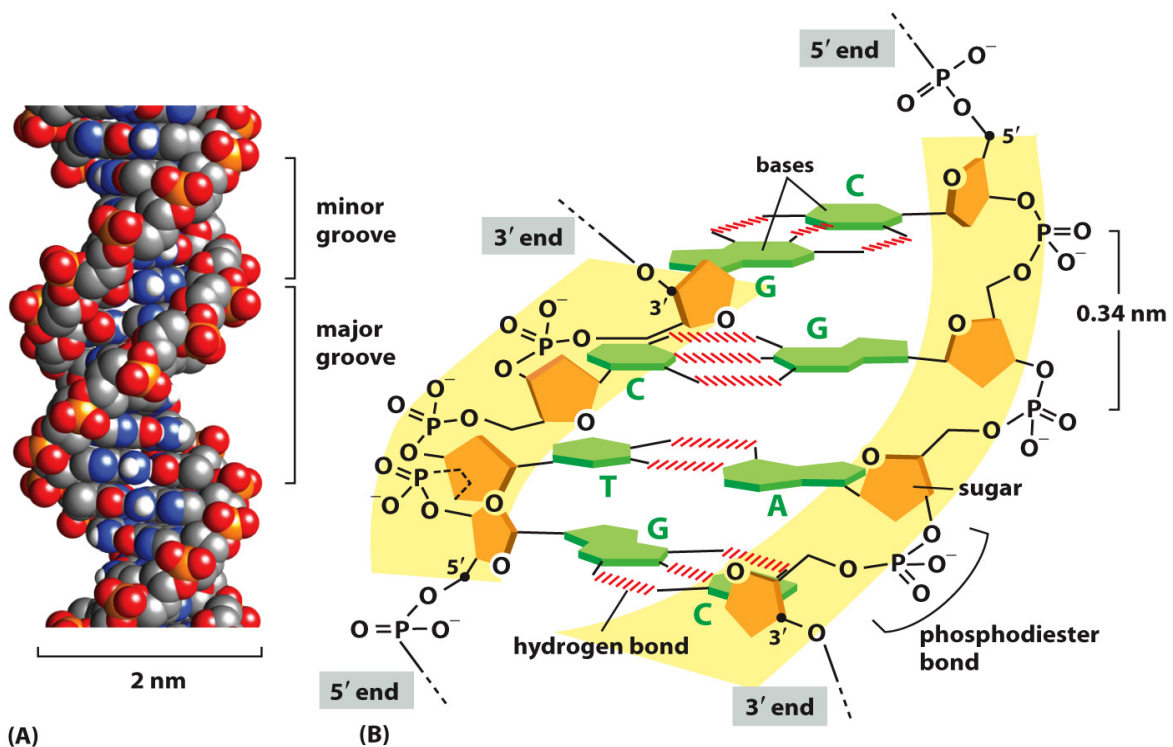
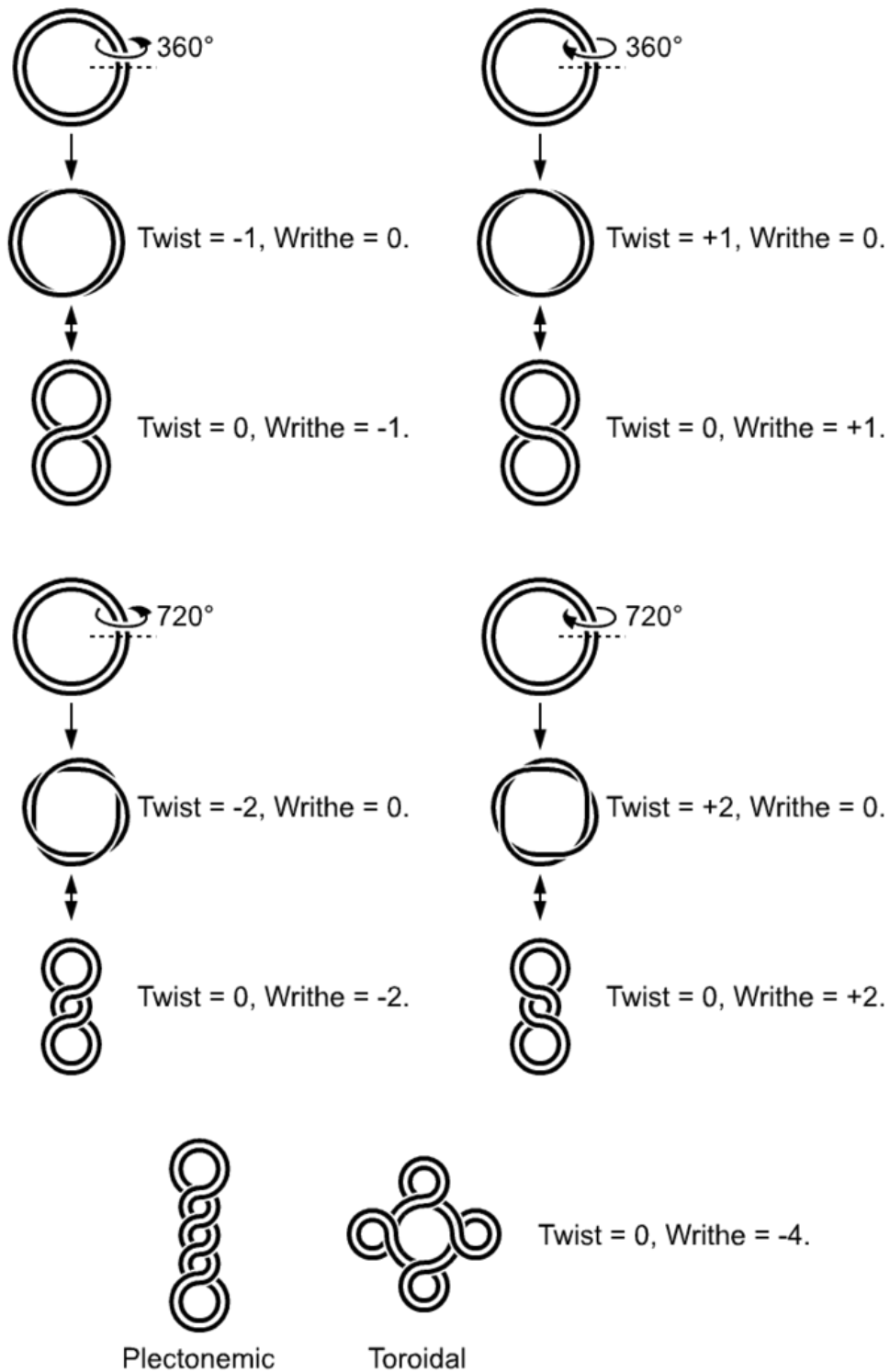


Figure 4-5 Molecular Biology of the Cell 6e (© Garland Science 2015)

Figure 4–5 The DNA double helix. (A) A space-filling model of 1.5 turns of the DNA double helix. Each turn of DNA is made up of 10.4 nucleotide pairs, and the center-to-center distance between adjacent nucleotide pairs is 0.34 nm. The coiling of the two strands around each other creates two grooves in the double helix: the wider groove is called the major groove, and the smaller the minor groove, as indicated.

(b) A short section of the double helix viewed from its side, showing four base pairs. The nucleotides are linked together covalently by phosphodiester bonds that join the 3'-hydroxyl (–OH) group of one sugar to the 5'-hydroxyl group of the next sugar. Thus, each polynucleotide strand has a chemical polarity; that is, its two ends are chemically different. The 5' end of the DNA polymer is by convention often illustrated carrying a phosphate group, while the 3' end is shown with a hydroxyl.



Supercoiling: Twist + Writhe = linking number. In this example the relaxed state is 0 coil.

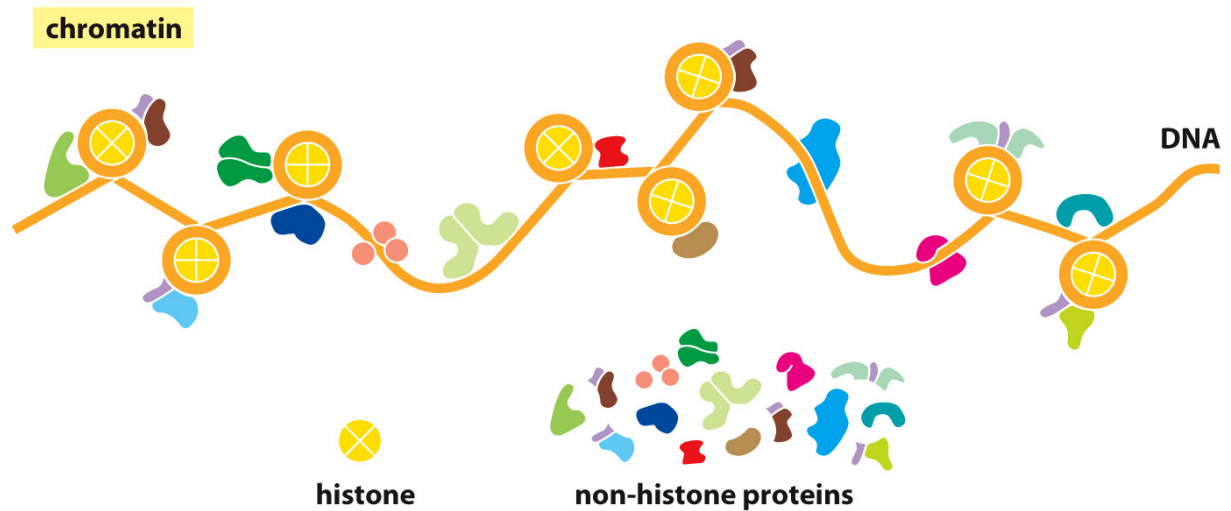


Figure 4-20 Molecular Biology of the Cell 6e (© Garland Science 2015)

Figure 4–20 Chromatin. As illustrated, chromatin consists of DNA bound to both histone and non-histone proteins. The mass of histone protein present is about equal to the total mass of non-histone protein, but—as schematically indicated here—the latter class is composed of an enormous number of different species. In total, a chromosome is about one-third DNA and two-thirds protein by mass

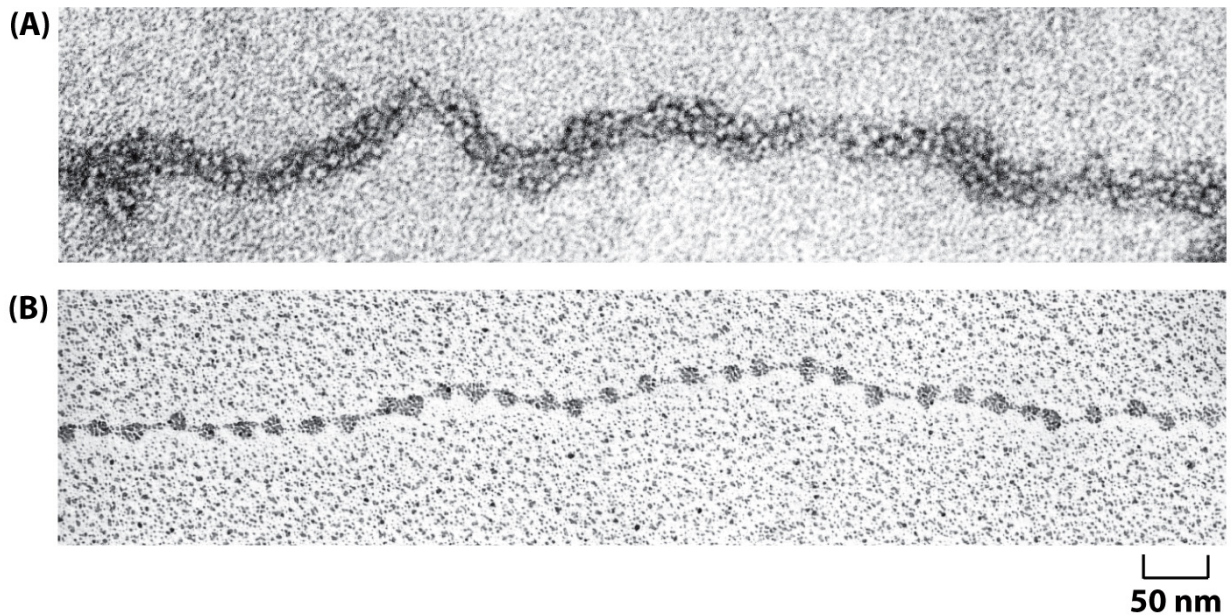


Figure 4-21 Molecular Biology of the Cell 6e (© Garland Science 2015)

Figure 4–21 Nucleosomes as seen in the electron microscope. (A) Chromatin isolated directly from an interphase nucleus appears in the electron microscope as a thread about 30 nm thick. (b) This electron micrograph shows a length of chromatin that has been experimentally unpacked, or decondensed, after isolation to show the nucleosomes. (A, courtesy of barbara Hamkalo; b, courtesy of victoria Foe.)

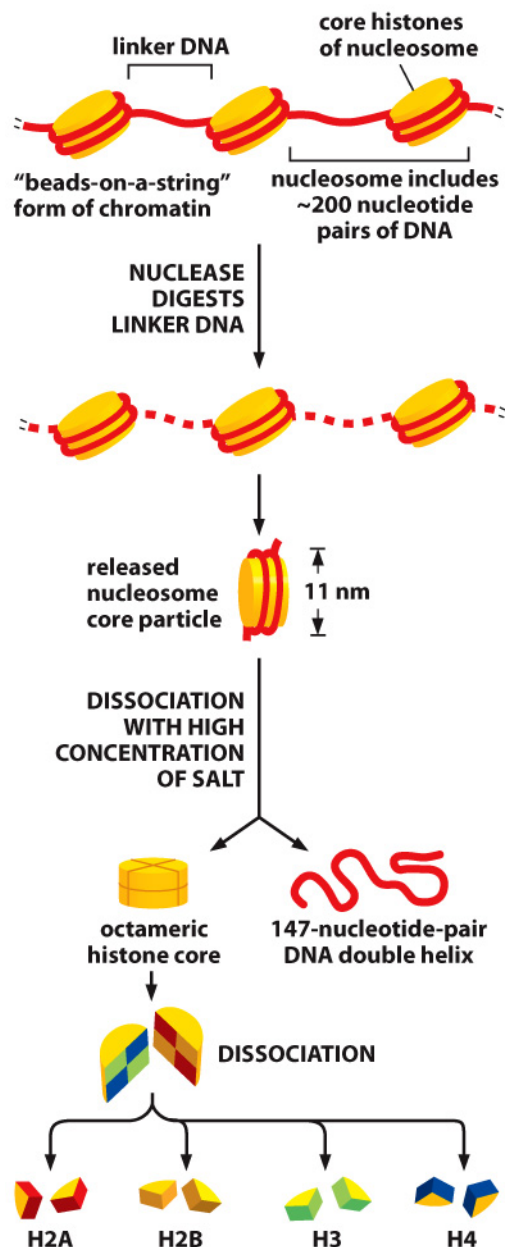


Figure 4–22 Structural organization of the nucleosome. A nucleosome contains a protein core made of eight histone molecules. In biochemical experiments, the nucleosome core particle can be released from isolated chromatin by digestion of the linker DNA with a nuclease, an enzyme that breaks down DNA. (The nuclease can degrade the exposed linker DNA but cannot attack the DNA wound tightly around the nucleosome core.) After dissociation of the isolated nucleosome into its protein core and DNA, the length of the DNA that was wound around the core can be determined. This length of 147 nucleotide pairs is sufficient to wrap 1.7 times around the histone core.

Figure 4-22 Molecular Biology of the Cell 6e (© Garland Science 2015)

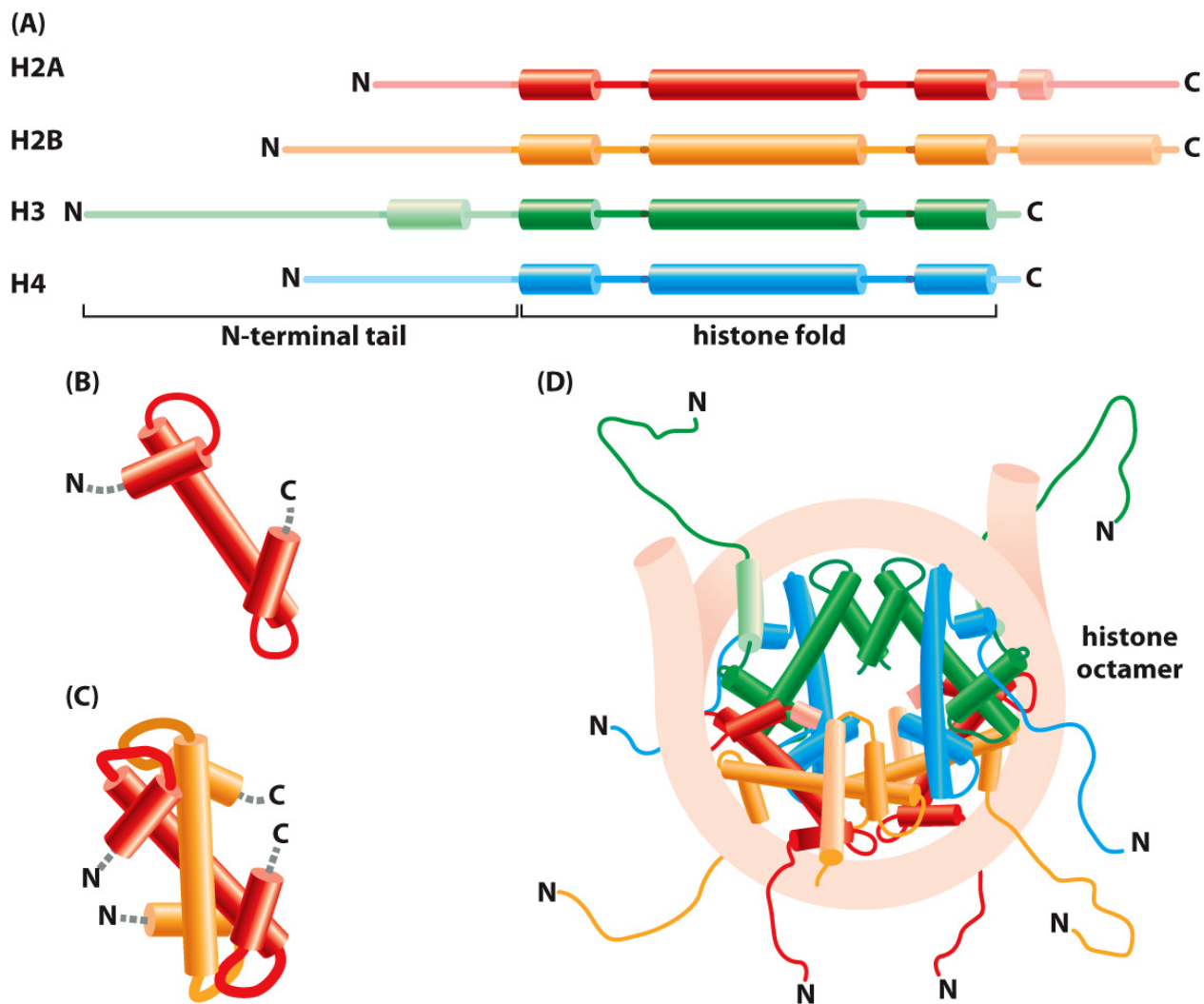


Figure 4-24 Molecular Biology of the Cell 6e (© Garland Science 2015)

Figure 4–24 The overall structural organization of the core histones. (A) Each of the core histones contains an N-terminal tail, which is subject to several forms of covalent modification, and a histone fold region, as indicated. (b) The structure of the histone fold, which is formed by all four of the core histones. (C) Histones 2A and 2b form a dimer through an interaction known as the “handshake.” Histones H3 and H4 form a dimer through the same type of interaction. (D) The final histone octamer on DNA. Note that all eight N-terminal tails of the histones protrude from the disc- shaped core structure. Their conformations are highly flexible, and they serve as binding sites for sets of other proteins.

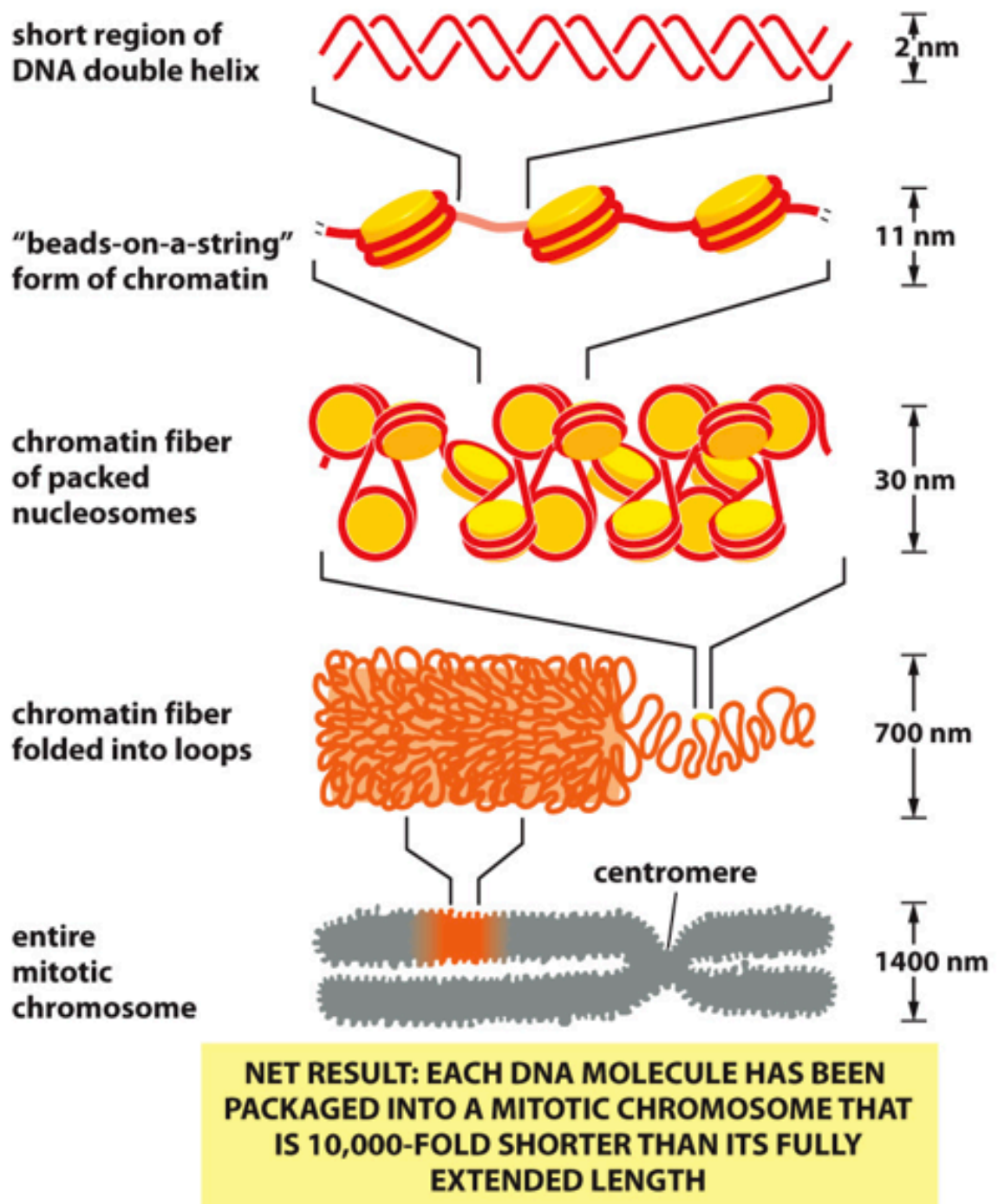


Figure 4-61 Molecular Biology of the Cell 6e (© Garland Science 2015)

Figure 4–61 Chromatin packing. This model shows some of the many levels of chromatin packing postulated to give rise to the highly condensed mitotic chromosome.